

# P vs NP as Conjecture-Grade UQFF Anchor: Informational Buoyancy Argument and Audit Closure

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## PAPER\_1193: P vs NP as Conjecture-Grade UQFF Anchor --- Informational Buoyancy Argument and Audit Closure

### Abstract

We close audit gap #11 of the UQFF calibration program by explicitly registering the P vs NP problem as a CONJECTURE-tier entry with zero observational anchors. The UQFF informational-buoyancy stance asserts that exhaustive search has positive vacuum (SCm) buoyancy cost,  $C_{\text{search}}(n) = B_{\text{inf}} \cdot 2^n$ , which diverges from any polynomial alternative  $C_{\text{solve}}(n) = B_{\text{inf}} \cdot n^k$  at  $n \rightarrow \infty$ , formally suggesting  $\text{P} \neq \text{NP}$ . We make this conjecture explicit and recordable in the calculator registry, but emphasize that no falsifiable observational anchor exists. Calculator registered as cp4\_id = 445; 20/20 smoke tests pass.

### 1. Motivation

Audit entry #11 noted that the UQFF framework had no formal handling of conjecture-grade mathematical claims with no observational anchor, breaking the G6 SM Anchor Gate audit trail. The fix is structural: explicitly mark such entries as CONJECTURE with anchor = None.

### 2. Informational Buoyancy Cost Functions

For an NP-complete decision problem of size  $n$ :

$C_{\text{search}}(n) \geq B_{\text{inf}} \cdot 2^n$ ,  $C_{\text{solve}}(n) \leq B_{\text{inf}} \cdot n^k$ ,  $k \in \mathbb{N}$ .

UQFF buoyancy non-negativity requires

$\boxed{\Delta_C(n) \equiv C_{\text{search}}(n) - C_{\text{solve}}(n) \geq B_{\text{inf}}(2^n - n^k)}$   
 $\forall n \geq n_*$

By contrapositive (informal): if  $\text{P} = \text{NP}$  existed then  $\Delta_C \rightarrow 0$  for all  $n$ , in tension with the asserted positivity of  $C_{\text{search}}$ . Hence the UQFF conjectural stance is  $\text{P} \neq \text{NP}$ .

{This argument is conjectural and has no observational anchor.}

### 3. Statement Registry

Four formal statements are exposed in STATEMENTS:

| ID | Name                            | Grade      | Anchor | UQFF Position                                                |
|----|---------------------------------|------------|--------|--------------------------------------------------------------|
| S1 | $\text{P} \neq \text{NP}$       | conjecture | None   | $\Delta_C > 0 \forall n \rightarrow \text{P} \neq \text{NP}$ |
| S2 | $\text{NP} \subseteq \text{AM}$ | conjecture | None   | orthogonal (no claim)                                        |

|    |                                      |            |      |                                        |
|----|--------------------------------------|------------|------|----------------------------------------|
| S3 | $\neg \text{co-NP}$                  | conjecture | None | proof/disproof informational asymmetry |
| S4 | NP-intermediate exists (Ladner 1975) | conjecture | None | consistent, not asserted               |

## 4. UQFF Aether Modulation

Even though the P vs NP problem is purely mathematical, the calculator returns a nominal aether factor  $f_A = 1 + \beta_i(\rho_{\text{SCm}}/\rho_{\text{amb}})\cos(\pi t_n)$  for registry-consistency; it does not affect the conjectural verdict.

## 5. Anchor Validation (Tier = CONJECTURE)

| Anchor | Type | Verdict                                                                 |
|--------|------|-------------------------------------------------------------------------|
| ---    | ---  | ---                                                                     |
| ---    | none | $n_{\text{anchors}} = 0$ ; $n_{\text{statements}} = 4$ ; all unanchored |

This is the correct outcome for a conjecture-tier entry. The calculator's `query_result` exposes `tier = "CONJECTURE"` so downstream audit tools can filter conjectural entries from observational anchor counts.

## 6. Code Registration

Module `_session301_pnp_conjecture_anchor.py`; class `PvsNPConjectureAnchor`; `cp4_id = 445`. Registered in `CondensedPhysics3.py` under `SESSION_301_CALCULATORS`. Smoke-test result: 20 / 20.

## 7. Falsifiable Predictions

- (i) Any future proof of  $P=NP$  would falsify the UQFF informational-buoyancy stance.
- (ii) The conjecture-tier flag must propagate unchanged in all downstream UQFF audit reports.

## 8. Status

$\text{[CLOSED]}$   $\quad$  audit gap  $\quad$   $\text{cp4\_id} = 445$   $\quad$  session = 301  $\quad$  tier = CONJECTURE.

## 9. References

Cook 1971; Karp 1972; Ladner 1975. Clay Mathematics Institute Millennium Prize Problems. UQFF\\_CALIBRATION\\_AUDIT.md.